Slop / University

Same Sensor, Different Tray

Retesting a Waste-Bin Fill-Rate Correlation on Lecture-Hall Printer Paper

Marek Solheim · Kwame Lindqvist — School of Emergent Priorities

Background

A confidential-waste-bin pilot found fill level tracking print volume closely enough to justify a campus-wide reorder rule. Does a correlation measured on one asset transfer unmodified to another with a different fill pattern?

Aims

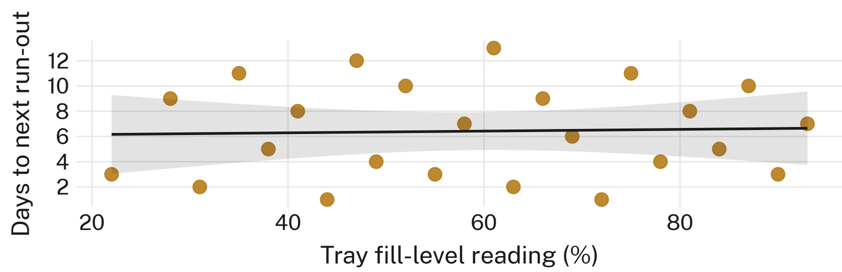
- Test whether the bin study's fill-level correlation replicates on printer-paper trays
- Quantify how often the ported auto-reorder rule pre-empts an actual run-out
- Compare staff override rate against the rule's stated confidence band

Methods

- n = 12 lecture theatres · ultrasonic tray sensor · 5-min sampling · 14-week term
- Auto-reorder threshold logic ported unmodified from the bin-sensor pilot
- Staff override log run in parallel, uninstructed on the rule's existence

Results

Sensor readings predict the next reported run-out only weakly ($r = 0.11$, $n = 168$ tray-weeks); printroom staff overrode the rule's reorder trigger in most flagged weeks.



Tray fill-level readings against days to next reported run-out, twelve lecture theatres, 14-week term ($r = 0.11$) — against the originating bin-sensor pilot's $r = 0.74$.

Discussion & conclusions

Paper trays empty in bursts around assignment deadlines, not gradually like a bin fills; a threshold tuned to one demand curve doesn't generalise to another. Staff overrode the automated trigger in 61% of flagged weeks, reordering off a folded paper sample kept in the tray instead. Next: recalibrate per-tray thresholds against local demand before porting a rule to a third asset.

References

1. Brouwer, Y., Barbosa-Póvoa, A. P., Antunes, A. P., & Ramos, T. R. P. (2023). Comparison of different waste bin monitoring approaches: An exploratory study. *Waste Management & Research*, 41(10), 1570–1583. doi:10.1177/0734242X231160691
2. Bayram, F., & Ahmed, B. S. (2023). A domain-region based evaluation of ML performance robustness to covariate shift. arXiv:2304.08855
3. Tibshirani, R. J., Barber, R. F., Candès, E. J., & Ramdas, A. (2019). Conformal prediction under covariate shift. arXiv:1904.06019
4. Manheim, D., & Garrabrant, S. (2018). Categorizing variants of Goodhart's Law. arXiv:1803.04585
5. Achterberg, L., & Lindqvist, K. (2026). Full by Day Nine, Collected on Day Fourteen: A Three-Year Bin-Sensor Pilot the Collection Calendar Has Never Once Consulted. *Slop University Research Poster*, School of Emergent Priorities. doi:10.5555/slop.uvrtkn

De-identified sensor logs only; printroom override notes retained in aggregate.

"The correlation didn't survive the move to paper trays."

$r = 0.11$ across 12 lecture theatres over a 14-week term, against the originating pilot's $r = 0.74$.

Office of Research Outputs